

MATLAB Script – Set 3

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Set 3:

A finite uniform linear charge, $\rho_L = 4 \frac{nC}{m}$, lies on the xy plane as shown in Figure 3.4, while point charges of 8 nC each are located at $(0, 1, 1)$ and $(0, -1, 1)$. Find $\vec{E}$ at $(0, 0, 0)$. Write a MATLAB program to verify your answer.

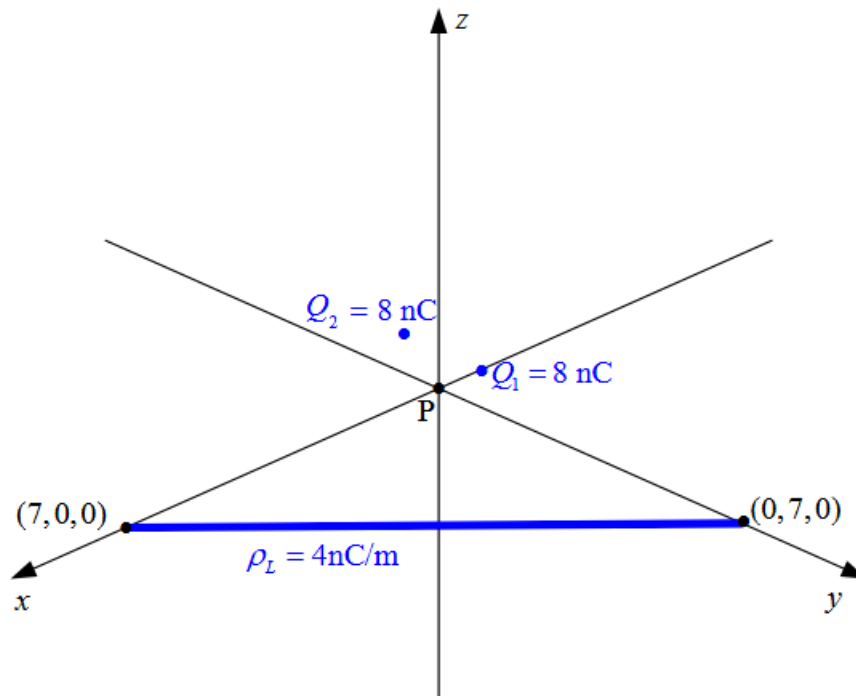


Figure 3.4 The charges of the exercise of Set 3.

Analytical Result:

Based on superposition, the electric field at $P(0, 0, 0)$ is $\vec{E} = \vec{E}_1 + \vec{E}_2 + \vec{E}_L$ where $\vec{E}_1$ and $\vec{E}_2$ are the electric fields generated by the point charges 1 and 2, respectively, and $\vec{E}_L$ is the electric field generated by the line charge.

Finding $\vec{E}_1$ (electric field generated by point charge Q_1):

Let $\vec{R}$ represent the vector pointing from the point charge to the observation point

$$P(0, 0, 0)$$

$$\text{Given: } Q_1 = 8 \text{ nC}, \epsilon_0 = 8.85 \times 10^{-12} \frac{C^2}{Nm^2}, \vec{R}_1 = a_y + a_z, \vec{R}_P = 0, \vec{R} = ?$$

$$\vec{R} = \vec{R}_P - \vec{R}_1 = 0 - (a_y + a_z) = -a_y - a_z$$

$$|\vec{R}| = \sqrt{(1)^2 + (1)^2} = \sqrt{1+1} = \sqrt{2}$$

$$\vec{E}_{point} = \frac{Q}{4\pi\epsilon_o|\vec{R}|^3} \vec{R}$$

$$\vec{E}_1 = \frac{Q_1}{4\pi\epsilon_o|\vec{R}|^3} \vec{R}$$

$$\vec{E}_1 = \frac{8 \times 10^{-9} C}{4\pi \left(8.85 \times 10^{-12} \frac{C^2}{Nm^2} \right) (\sqrt{2})^3} (-a_y - a_z)$$

$$\vec{E}_1 \approx -25.433a_y - 25.433a_z \frac{N}{C}$$

Finding $\vec{E}_2$ (electric field generated by point charge Q_2):

Let $\vec{R}$ represent the vector pointing from the point charge to the observation point

$$P(0, 0, 0)$$

$$\text{Given: } Q_2 = 8 \text{ nC}, \epsilon_o = 8.85 \times 10^{-12} \frac{C^2}{Nm^2}, \vec{R}_2 = -a_y + a_z, \vec{R}_P = 0, \vec{R} = ?$$

$$\vec{R} = \vec{R}_P - \vec{R}_2 = 0 - (-a_y + a_z) = a_y - a_z$$

$$|\vec{R}| = \sqrt{(1)^2 + (1)^2} = \sqrt{1+1} = \sqrt{2}$$

$$\vec{E}_{point} = \frac{Q}{4\pi\epsilon_o|\vec{R}|^3} \vec{R}$$

$$\vec{E}_2 = \frac{Q_2}{4\pi\epsilon_o|\vec{R}|^3} \vec{R}$$

$$\vec{E}_2 = \frac{8 \times 10^{-9} C}{4\pi \left(8.85 \times 10^{-12} \frac{C^2}{Nm^2} \right) (\sqrt{2})^3} (a_y - a_z)$$

$$\vec{E}_2 \approx 25.433a_y - 25.433a_z \frac{N}{C}$$

Finding $\vec{E}_L$ (electric field generated by linear charge ρ_L):

$$\text{Given: } \epsilon_o = 8.85 \times 10^{-12} \frac{C^2}{Nm^2}, \rho_L = 4 \times 10^{-9} C/m$$

Choose convenient coordinate along the line:

$$y = 7 - x, 0 \leq x \leq 7, z = 0$$

Small segment length:

$$dl = \sqrt{dx^2 + dy^2} = \sqrt{dx^2 + (-dx)^2} = \sqrt{2}dx$$

Charge element:

$$dq = \rho_L dl = \rho_L \sqrt{2}dx$$

Distance from dq to origin:

$$\text{Source point: } (x, 7 - x, 0)$$

$$R = \sqrt{x^2 + (7 - x)^2}$$

$$R^2 = x^2 + (7 - x)^2$$

$$R^3 = (x^2 + (7 - x)^2)^{\frac{3}{2}}$$

Field magnitude from dq:

$$dE = \frac{dq}{4\pi\epsilon_o R^2} = \frac{\rho_L \sqrt{2}dx}{4\pi\epsilon_o (x^2 + (7 - x)^2)}$$

Direction is from charge toward origin, proportional to $-(x, 7 - x, 0)$

$$dE_x = dE \left(-\frac{x}{R} \right)$$

$$dE_y = dE \left(\frac{-(7 - x)}{R} \right)$$

$$dE_z = 0$$

Substitute dE and R:

$$dE_x = \frac{-x\rho_L\sqrt{2}}{4\pi\epsilon_o (x^2 + (7 - x)^2)^{\frac{3}{2}}} dx$$

$$dE_y = \frac{-(7 - x)\rho_L\sqrt{2}}{4\pi\epsilon_o (x^2 + (7 - x)^2)^{\frac{3}{2}}} dx$$

Integrate $x = 0$ to 7

$$dE_x = \frac{-\rho_L\sqrt{2}}{4\pi\epsilon_o} \int_0^7 \frac{x}{(x^2 + (7 - x)^2)^{\frac{3}{2}}} dx$$

$$dE_y = \frac{-\rho_L \sqrt{2}}{4\pi\epsilon_o} \int_0^7 \frac{7-x}{(x^2 + (7-x)^2)^{\frac{3}{2}}} dx$$

Symmetry:

If $u = 7 - x$, $E_x = E_y$. Thus compute one integral, call it I , then $E_x = E_y = \frac{-\rho_L \sqrt{2}}{4\pi\epsilon_o} I$

Evaluate Integral:

Simplify inside:

$$x^2 + (7-x)^2 = 2x^2 - 14x + 49 = 2[(x-3.5)^2 + (3.5)^2]$$

$$\text{Let } t = x - 3.5, a = 3.5$$

$$\text{Then } x = t + a, dx = dt$$

$$x^2 + (7-x)^2 = 2(t^2 + a^2)$$

$$(x^2 + (7-x)^2)^{\frac{3}{2}} = (2(t^2 + a^2))^{\frac{3}{2}} = 2\sqrt{2}(t^2 + a^2)^{\frac{3}{2}}$$

Integral:

$$I = \int_0^7 \frac{7-x}{(x^2 + (7-x)^2)^{\frac{3}{2}}} dx = \int_{-a}^a \frac{t+a}{2\sqrt{2}(t^2 + a^2)^{\frac{3}{2}}} dt$$

Split Integral:

Odd part:

$$\int_{-a}^a \frac{t}{2\sqrt{2}(t^2 + a^2)^{\frac{3}{2}}} dt = 0$$

Even part remains:

$$I = \frac{a}{2\sqrt{2}} \int_{-a}^a \frac{1}{(t^2 + a^2)^{\frac{3}{2}}} dt$$

Known result:

$$\int \frac{1}{(t^2 + a^2)^{\frac{3}{2}}} dt = \frac{t}{a^2 \sqrt{t^2 + a^2}}$$

So:

$$\int_{-a}^a \frac{1}{(t^2 + a^2)^{\frac{3}{2}}} dt = \left. \frac{t}{a^2 \sqrt{t^2 + a^2}} \right|_{-a}^a = \frac{2a}{a^2 \sqrt{2a^2}} = \frac{2a}{a^2 (a\sqrt{2})} = \frac{2}{a^2 \sqrt{2}}$$

Thus:

$$I = \frac{a}{2\sqrt{2}} \times \frac{2}{a^2\sqrt{2}} = \frac{1}{2a}$$

With $a = 3.5$:

$$I = \frac{1}{7}$$

Final Line Charge:

$$E_x = E_y = \frac{-\rho_L \sqrt{2}}{4\pi\epsilon_0} I = \frac{-(4 \times 10^{-9} \text{ C/m})\sqrt{2}}{4\pi(8.85 \times 10^{-12} \frac{\text{C}^2}{\text{Nm}^2})} \times \frac{1}{7} \approx -7.27$$

Vector:

$$\vec{E}_L = -7.27(a_x + a_y)$$

$$\therefore \vec{E}_L = -7.27a_x - 7.27a_y \frac{\text{N}}{\text{C}}$$

Calculating $\vec{E}$ (Total Field):

$$\vec{E} = \vec{E}_1 + \vec{E}_2 + \vec{E}_L$$

$$\vec{E} = (-25.433a_y - 25.433a_z) \frac{\text{N}}{\text{C}} + (25.433a_y - 25.433a_z) \frac{\text{N}}{\text{C}} + (-7.27a_x - 7.27a_y) \frac{\text{N}}{\text{C}}$$

$$\vec{E} \approx -7.27a_x - 7.27a_y - 50.87 \frac{\text{N}}{\text{C}}$$

$\therefore$ the total electric field at $P(0,0,0)$ is approximately $-7.27a_x - 7.27a_y - 50.87 \frac{\text{N}}{\text{C}}$

which was close to the result the MATLAB code generated verifying our results.

The MATLAB code generated an answer of $-7.266a_x - 7.266a_y - 50.865 \frac{\text{N}}{\text{C}}$

To generate a closer result, the discretization step size for the integration in the MATLAB code would simply be decreased which would increase accuracy while also increasing computational expense.

MATLAB Output to Verify Answers:

Command Window

```
The electric field at point P(0,0,0) is [-7.266, -7.266, -50.865] N/m  
fx >>
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